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**Project Based Learning Report
On**

**“Region Splitting and Merging for Terrain Classification using
Quadtree Segmentation”**

Subject: - Image Processing Semester/Year: III/V

Sr. No	Name of Students & USN	Problem Understanding & Objective[2M]	Implementation & Functionality[4 M]	Output Quality & Analysis[2 M]	Report/Presentation Quality[2 M]	Total[10M]
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Date of Submission:

1. ABSTRACT

Terrain classification is an important application of image processing and remote sensing in which an image is divided into meaningful regions such as vegetation, water bodies, mountains, barren land, agricultural areas, and urban regions. Since terrain images generally contain regions with different textures, intensities, and spatial characteristics, efficient image segmentation is required for accurate classification. This project studies and implements Region Splitting and Merging using Quadtree Segmentation for terrain classification. If a region is not homogeneous according to a predefined similarity criterion, it is divided into

four equal subregions. Once sufficiently homogeneous regions are obtained, adjacent or similar regions are merged to form larger meaningful terrain regions.

2. INTRODUCTION

Image segmentation is one of the fundamental operations in digital image processing. It involves dividing an image into multiple meaningful regions so that pixels belonging to the same object or area are grouped together. Effective segmentation is particularly important in terrain classification because natural scenes contain different land-cover categories with varying intensity, colour, texture, and spatial properties. Terrain classification involves identifying and categorizing different types of terrain from an image. **Region splitting and merging** is a region-based segmentation technique that addresses this problem by considering groups of pixels rather than individual pixels. In region splitting, the complete image is initially considered as one region. If the region does not satisfy a homogeneity criterion, it is split into smaller regions.

A **Quadtree** is particularly suitable for this process because every non-leaf node is divided into four equal-sized regions. Therefore, a large non-uniform terrain region can be recursively divided into smaller regions, while homogeneous areas can remain as larger blocks.

3. AIM AND OBJECTIVES

Aim: To implement and study Region Splitting and Merging using Quadtree Segmentation for effective segmentation and classification of different terrain regions in digital images

Objective:

1. To understand the concept of image segmentation and region-based processing.
2. To study the principles of region splitting and region merging.
3. To understand the structure and working of a Quadtree.

4. PROBLEM STATEMENT

Terrain images generally contain different geographical and land-cover regions with varying intensity, colour, and texture. Identifying these regions manually can be time-consuming, especially when the image is large or contains complex terrain patterns.

Traditional pixel-based classification considers individual pixels independently and may fail to capture the spatial relationship between neighbouring pixels. Similarly, applying a uniform segmentation strategy to the complete image may result in either excessive segmentation or loss of important details.

The problem addressed in this project is to segment a terrain image into meaningful homogeneous regions using Region Splitting and Merging with a Quadtree structure. Initially, the entire terrain image is considered as a single region. The region is evaluated using a predefined homogeneity criterion such as intensity variance, colour similarity, or texture similarity. If the region is not homogeneous, it is divided into four equal subregions. This process is repeated recursively.

5. LITERATURE REVIEW

Image segmentation has been extensively studied as an important stage in image processing and computer vision. Region-based segmentation methods classify pixels by considering their relationships with neighbouring pixels and the properties of complete regions.

Region growing is one of the basic region-based approaches in which pixels or small regions are combined when they satisfy a similarity criterion. The **split-and-merge technique** combines these two concepts. Splitting allows non-uniform regions to be subdivided, while merging combines similar neighbouring regions. This provides a more flexible segmentation strategy.

Quadtree representation is commonly associated with hierarchical spatial decomposition. In a Quadtree, an image region is divided into four equal parts whenever additional detail is required. In terrain and remote-sensing applications, segmentation can help distinguish different land-cover categories. However, terrain classification can be affected by illumination, shadows, seasonal changes, sensor characteristics, noise, and similarity between different terrain classes.

6. PROBLEM DEFINITION & METHODOLOGY

Problem Definition: The project considers a terrain image containing multiple regions such as vegetation, water, soil, roads, buildings, or mountainous areas. The objective is to divide the image into regions according to a predefined homogeneity criterion.

Methodology: The methodology begins by obtaining a suitable terrain image. The image may be converted into an appropriate colour space or grayscale representation depending on the selected similarity criterion. Initially, the complete image is considered as a single region. The statistical properties of the region are calculated.

- Top-left
- Top-right
- Bottom-left
- Bottom-right

Each subregion is then independently tested for homogeneity. Non-homogeneous subregions are recursively divided again. After the splitting stage, neighbouring regions are examined for similarity. Regions having similar mean intensity, colour, or other selected features are merged. The final result is a segmented terrain image containing meaningful regions.

7. MATHEMATICAL EQUATION/ MODELING

Let the input terrain image be represented by:

$$I(x, y)$$

where x and y represent the pixel coordinates. For a region R , calculate its mean intensity:

$$\mu_R = \frac{1}{N} \sum_{(x,y) \in R} I(x,y)$$

where N is the number of pixels in region R . The variance of the region is:

$$\sigma_R^2 = \frac{1}{N} \sum_{(x,y) \in R} (I(x,y) - \mu_R)^2$$

A homogeneity condition can be defined as:

$$\sigma_R^2 \leq T$$

where T is a predefined threshold.

Splitting condition

If:

$$\sigma_R^2 > T \text{ then region } R \text{ is considered non-}$$

homogeneous and is split into four subregions:

$$R \rightarrow R_1 \cup R_2 \cup R_3 \cup R_4$$

If:

$$\sigma_R^2 \leq T$$

the region is considered homogeneous and no further splitting is required.

Merging condition

Two adjacent regions R_i and R_j can be merged if their characteristics are sufficiently similar:

$$|\mu_{R_i} - \mu_{R_j}| \leq T_m$$

where T_m is the merging threshold.

8. PSEUDOCODE/ALGORITHM

PseudoCode:

START

1. Upload terrain image.
2. Resize image to processing size.
3. Create ROOT region containing the whole image.
4. For each region:
Calculate mean intensity and variance.

5. If variance > threshold AND depth < maximum depth: Split region into 4 quadrants.
Repeat Step 4 for each quadrant.
Else:
Store region as a leaf region.
6. Compare neighboring leaf regions.
7. If neighboring regions have similar intensity and terrain class:
Merge them.
8. Calculate average RGB values of each final region.
9. Classify each region as:
Water / Forest / Soil / Rocky / Urban.
10. Display:
Original Image
Quadtree Regions
Merged Regions
Classified Image
Statistics/Histogram

END

9. IMPLEMENTATION

The project can be implemented using Python and image-processing libraries such as OpenCV and NumPy.

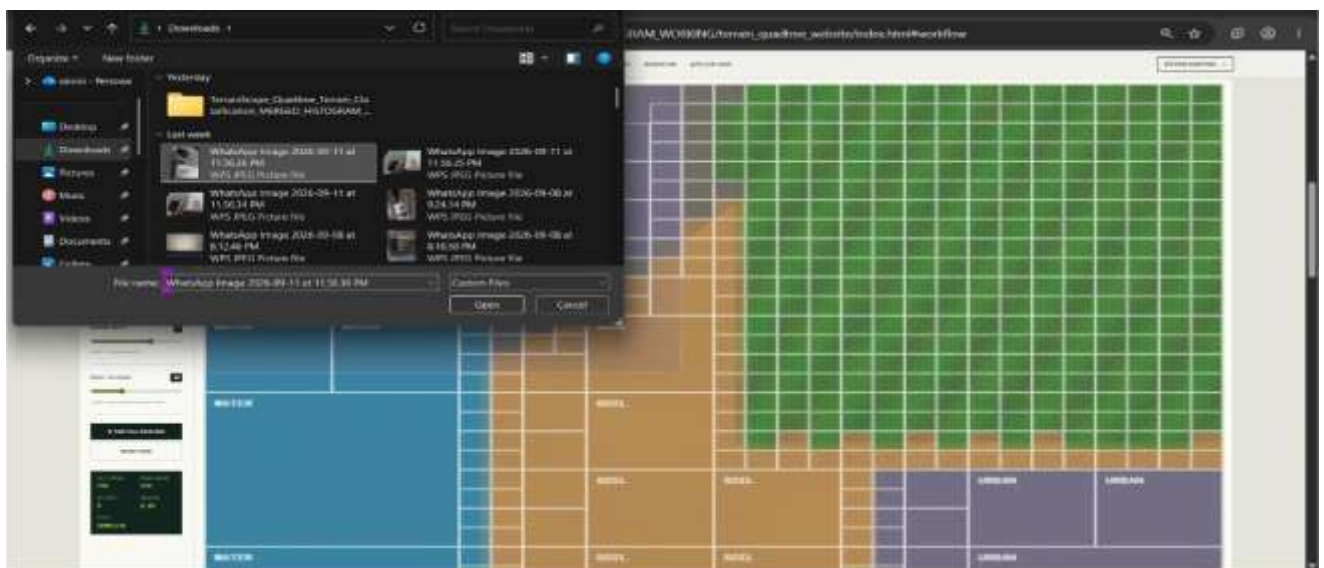
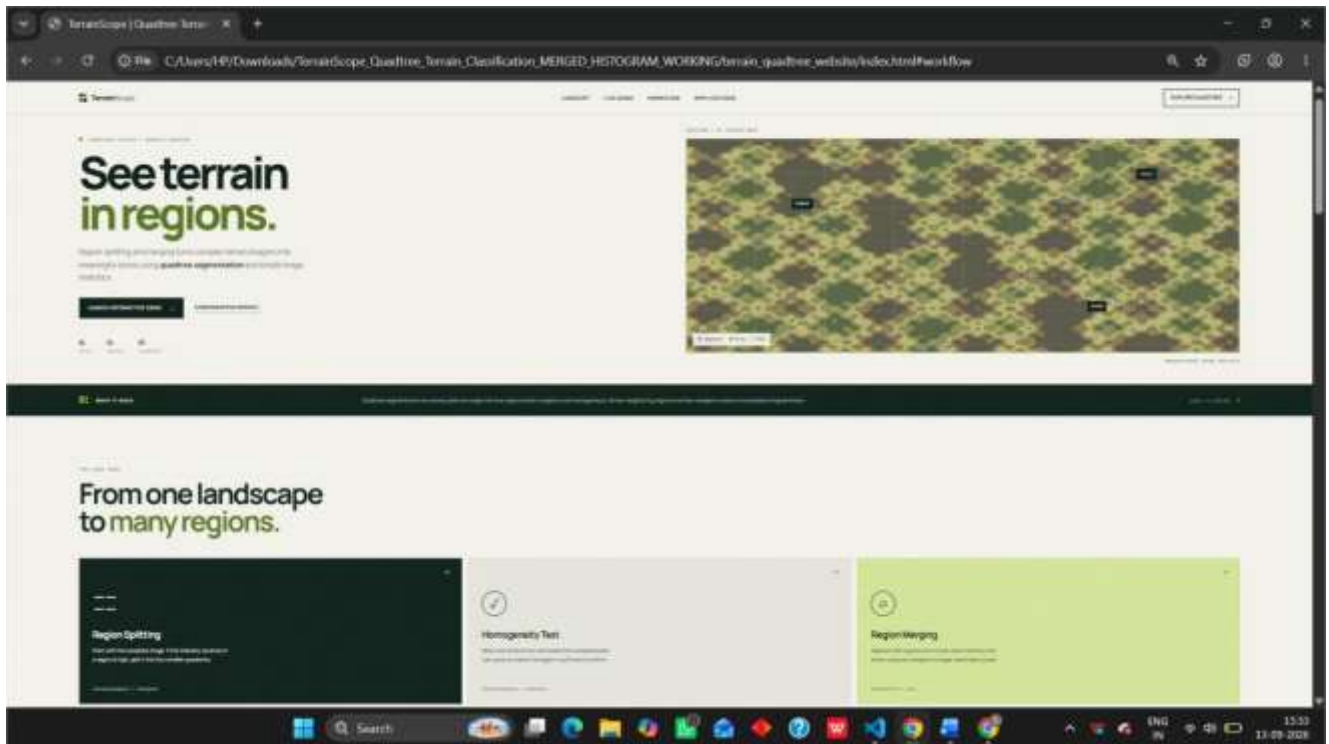
- 1. Image Input:** The terrain image is loaded using an image-processing library.
- 2. Preprocessing :** The image can be resized and converted into grayscale or another suitable representation. Noise reduction may also be performed when required.
- 3. Homogeneity Measurement:** For each region, statistical properties such as mean and variance are calculated.

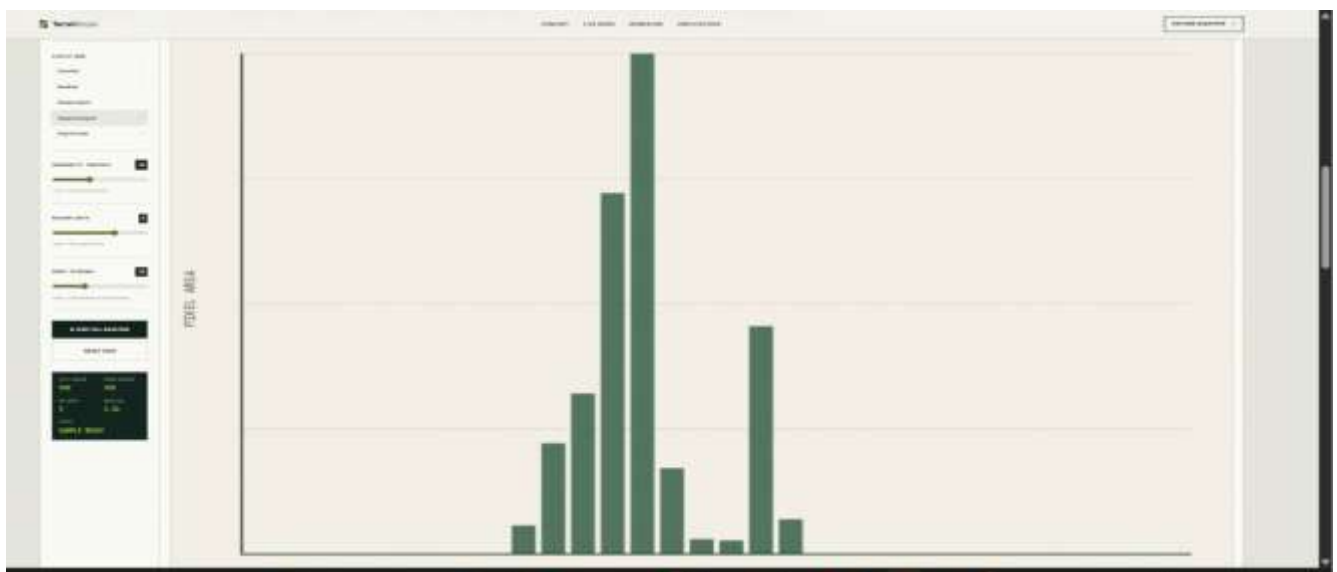
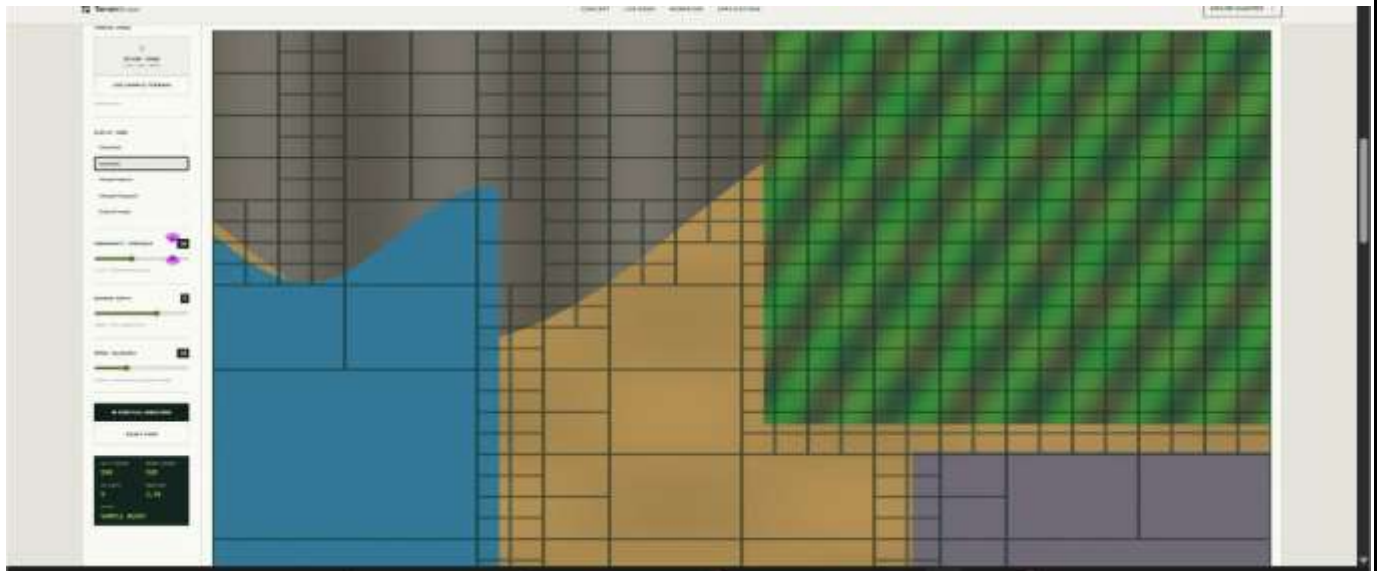
10. RESULTS AND DISCUSSION

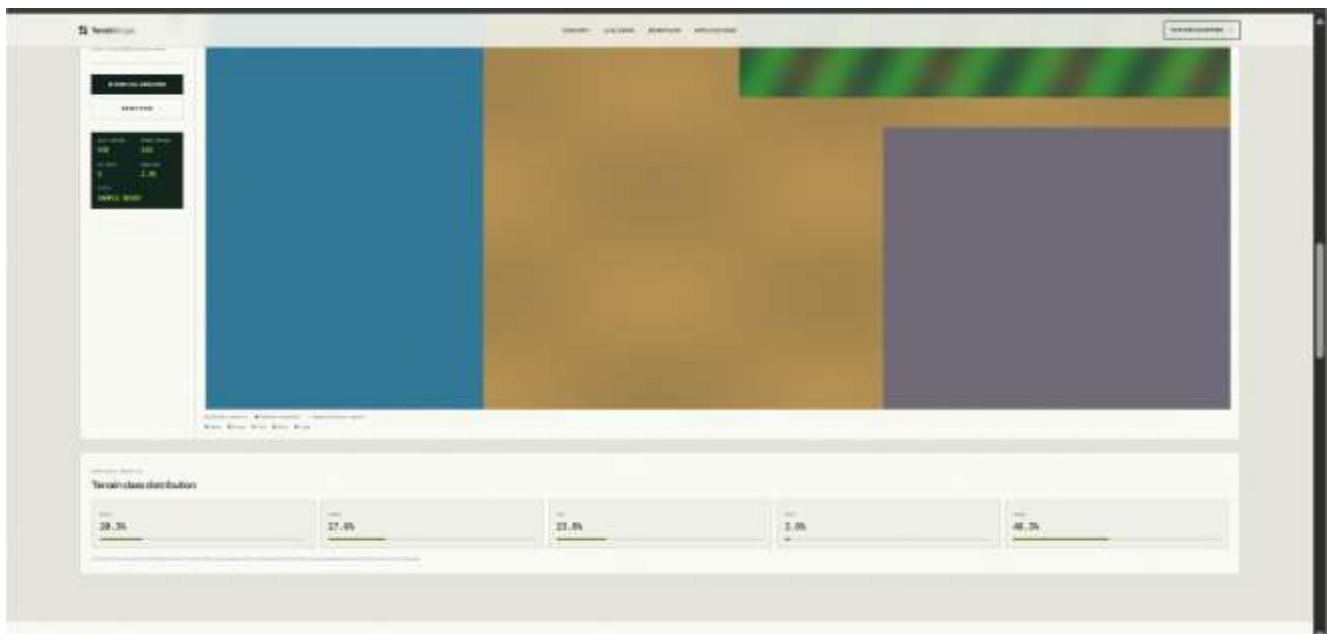
Result

Parameter	Observation
Input	Terrain image
Initial Region	Complete image
Segmentation Method	Region Splitting and Merging

Output Images:







11. CONCLUSION AND FUTURE SCOPE

Conclusion: Region Splitting and Merging using Quadtree Segmentation provides an effective region-based approach for segmenting terrain images. The method begins by treating the entire image as one region and recursively splitting non-homogeneous regions into four smaller regions.

Future Scope: The project can be extended by incorporating advanced features such as: Colour-based segmentation for improved terrain classification. Texture analysis for distinguishing visually similar terrain regions.

12. REFERENCES

1. Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*, Pearson.
2. S. Sridhar, *Digital Image Processing*, Oxford University Press.
3. Milan Sonka, Vaclav Hlavac, and Roger Boyle, *Image Processing, Analysis, and Machine Vision*, Cengage Learning.
4. David A. Forsyth and Jean Ponce, *Computer Vision: A Modern Approach*, Pearson.
5. OpenCV Documentation – Image Processing and Segmentation.

13. APPENDIX

A. Suggested Python Implementation

The implementation can use:

- **Python** – Programming language • **OpenCV** – Image processing
- **NumPy** – Numerical calculations
- **Matplotlib** – Visualization
- **Jupyter Notebook / VS Code** – Development environment

A basic implementation can follow this structure: import

```
cv2 import numpy as np import matplotlib.pyplot as plt
```

```
image = cv2.imread("terrain.jpg") gray =
```

```
cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
threshold = 20 min_size = 16
```

```
regions = []
def split_region(img, x, y, w, h):
    region = img[y:y+h, x:x+w]
    variance = np.var(region)
    if variance <= threshold or w <= min_size or h <= min_size:
        regions.append((x, y, w, h))
    return w2 = w // 2, h2 = h // 2
    split_region(img, x, y, w2, h2)
    split_region(img, x+w2, y, w-w2, h2)
    split_region(img, x, y+h2, w2, h-h2)
    split_region(img, x+w2, y+h2, w-w2, h-h2)
h, w = gray.shape
split_region(gray, 0, 0, w, h)
output = image.copy()
for x, y, rw, rh in regions:
    cv2.rectangle(output, (x, y), (x+rw, y+rh), (0, 255, 0), 1)

plt.figure(figsize=(10, 6))

plt.imshow(cv2.cvtColor(output, cv2.COLOR_BGR2RGB))
plt.title("Quadtree Terrain Segmentation")
plt.axis("off")
plt.show()
```

B. Applications

The implemented Quadtree segmentation method can be used as a preprocessing technique for terrain classification. The segmented regions can subsequently be classified into categories such as vegetation, water bodies, agricultural land, barren land, urban regions, and mountainous areas based on their intensity, colour, texture, or other image feature.